



# Climate Variability and Household Electricity Consumption

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**Abstract:** This study aims to inform evidence-based policy for climate change adaptation in the Philippines by estimating the climate sensitivity of household electricity demand among select cities in Metro Manila during the COVID-19 pandemic. Localized climate data from 2001 to 2021 were matched to households based on their city of residence to construct measures of climate variability. Heat Index (HI), which combines temperature and relative humidity, provides a measure of perceived thermal discomfort. Cooling Degree Days (CDD), defined as the number of days when daily HI exceeds a long-run threshold of 34.82 °C (the 75th percentile of historical HI), serve as another indicator of exposure to extreme heat conditions. Correlated random effects (Mundlak) regressions reveal strong climate sensitivity of electricity use. Household electricity consumption increases when high temperature coincides with high humidity. A 1 °C rise in HI leads to a 19 kWh increase in monthly electricity use, while an additional CDD raises monthly consumption by about 6 kWh. The estimated positive relationship between temperature, humidity, and electricity consumption is consistent with increased demand for cooling services during periods of higher thermal stress. Overall, the results suggest that households adjust their electricity consumption in response to perceived heat conditions through a range of cooling-related behaviors. However, electricity demand among lower-income households is significantly less responsive to climate fluctuations, which may indicate limited adaptive capacity amidst greater thermal discomfort. Conversely, higher-income households exhibit greater responsiveness, highlighting their potential for targeted adoption of energy-efficient appliances and rooftop solar systems. The results underscore the importance of climate-sensitive energy policies that simultaneously advance climate mitigation goals and address equity concerns.

**Keywords:** Electricity consumption; Climate; Temperature; Relative humidity; Heat Index; Correlated random effects (Mundlak) regression; Equity concerns

## 1. Introduction

Global warming, defined as the long-term rise in the Earth's surface temperature since the Industrial Revolution, has intensified markedly in the twenty-first century (Intergovernmental Panel on Climate Change, 2021). The IPCC projected that the average global surface temperature would increase by 2.6 to 4.8 °C by the end of the century (Intergovernmental Panel on Climate Change, 2013). The World Meteorological Organization (2022) reported that the years 2015 to 2021 were the seven warmest years on record, with global mean temperature in 2021 being  $1.11 \pm 0.13$  °C above the pre-industrial (1850 to 1900) average. In just three years, World Meteorological Organization (2025) reports that global average surface temperature in 2024 already reached  $1.55 \pm 0.13$  °C above the pre-industrial average, likewise making the last 10 years the warmest years on record.

The rise in surface temperature has been accompanied by notable increases in the frequency and intensity of extreme heat events, contributing substantially to growing electricity demand (Schaeffer et al., 2012). Further, the effect of temperature on electricity demand has been shown to increase in importance in recent years. Using multi-country, time-series data, Lee & Chiu (2011) found that over time, the elasticity of electricity demand with respect to temperature gradually increases.

Moral-Carcedo & Pérez-García (2015) observed that electricity demand in the residential and commercial sectors responds more strongly to high temperatures, making these sectors the main contributors to peak electricity consumption during hotter months. Bin & Dowlatabadi (2005) theorized that a household's decision on how much

energy to use during extreme weather events is largely influenced by considerations of comfort maintenance and health protection. Empirically, substantial increases in household electricity demand due to more frequent use of air-conditioners during the hot seasons have been found in Thailand (Arifwidodo & Chandrasiri, 2015), Hong Kong (Lam et al., 2008), China (Li et al., 2019), Ghana (Avordeh et al., 2021), Cyprus (Zachariadis & Pashourtidou, 2007), and the United States (Mansur et al., 2008).

In the Philippines, a rapidly developing country located in tropical Southeast Asia, the residential sector is a major contributor to electricity demand. The Philippine Energy Plan 2018–2040 reports that household sector tops both the industrial and commercial sectors in terms of electricity consumption. For on-grid electricity sales, the residential sector's 36.2% share was trailed by the commercial sector's 31.1% and industrial sector's 30.5% shares. For off-grid electricity sales, the residential sector's accounted for the majority share at 55.9% (Department of Energy, 2021). Thus, the household sector has a key role in climate change mitigation and adaptation (Palanca-Tan, 2024).

Estimates of the impact of climate change on household energy demand are typically used to predict the cost of climate change adaptation (Auffhammer & Mansur, 2014). Hence, these estimates provide essential inputs in the formulation and implementation of mitigation and adaptation policies, such as the promotion of energy-saving cooling technologies and roof-top solar power for households.

This paper aims to contribute to evidence-based policy making and program development for climate change mitigation and adaptation in the Philippines by estimating household electricity-demand response functions to climate variability. The specific tasks undertaken in this research are as follows: (1) Derive measures of climate variability in Metro Manila, Philippines; (2) Estimate the effect of climate variability on monthly electricity consumption of Metro Manila households; (3) Identify other factors such as household characteristics and appliance usage that contribute to household electricity consumption; and (4) Derive the climate-electricity demand response functions across low, middle, and high household income groups.

To the authors' knowledge, this is the first study on the link between climate and electricity demand using monthly household-level data in the Philippines. Existing empirical estimates on the climate-household electricity demand response function are still largely concentrated in Western countries (Li et al., 2019). In Southeast Asia, existing literature has focused on appliance usage, house type, household characteristics and government support as determinants of electricity demand (Akil et al., 2021; Ali et al., 2020). A study in Thailand investigated the impact of the urban heat island development on household energy consumption (Arifwidodo & Chandrasiri, 2015). This paper adds to the limited literature on climate change and household energy demand in developing economies. Further, the paper looks at relative climate-electricity demand sensitivities among different household income groups. While macro-level studies indicating an increase in electricity demand sensitivity to higher temperature with income growth abound, household-level studies are just starting to emerge.

## 2. Methodology

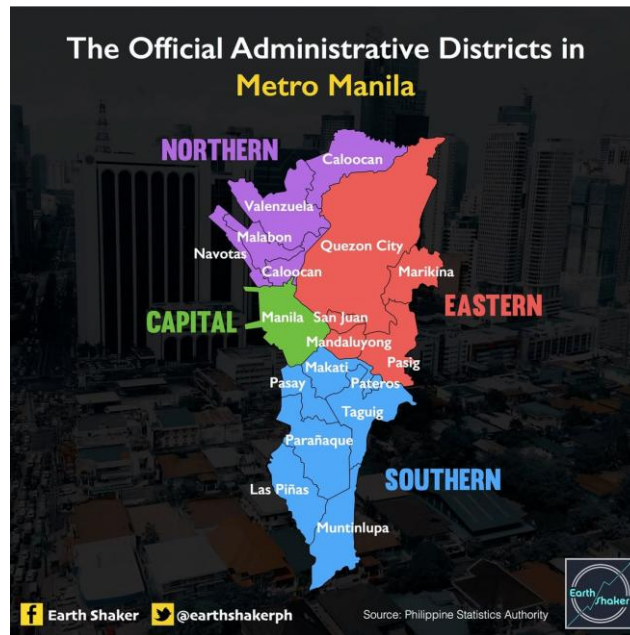
### 2.1 Data

Two sets of data were used for this study, household electricity consumption survey data and climatological data.

#### 2.1.1 Household survey

A household survey was conducted for this study between December 2020 and May 2021. The household's monthly electricity consumption volume (in kWh) in the last 12 months and the peso value of the latest monthly electricity consumption were obtained from the household's latest electricity bill. Additional data collected in the survey include monthly household income, household composition (total number of household members, number of household members staying at home at daytime), and household's electrical appliances.

Survey respondents were households residing in the Philippines' National Capital Region (NCR), which is also the country's political, economic, social and cultural center. While it is the smallest among the country's 17 regions (area of only 620 km<sup>2</sup>), NCR is the second most populous (13.5 million in 2020, 12.4% of the entire Philippine population), and the most densely populated (21,749 people per km<sup>2</sup> in 2020) (Philippine Statistics Authority, 2021). The study employed multi-staged stratified sampling procedure to ensure coverage across major geographic areas of the NCR. The four NCR districts comprised the first-stage strata, while the cities in each district comprised the second-stage strata. The principal city from each district was selected as representative city: Manila City for the Capitol District; Quezon City for the Eastern District; Caloocan City for the Northern District; and Makati City for the Southern District (Figure 1). Each sampled household must have its own metered connection to the Manila Electric Company (MERALCO), the largest electric distribution utility company in the Philippines that accounts for 55% of the country's total electricity distribution (Manila Electric Company, 2024).



**Figure 1.** Survey site  
Source: Earth Shaker (2019)

A total sample of 500 respondents was identified from predominantly residential areas with a mix of low-, middle-, and high-income households. Respondents from each survey area were chosen using systematic sampling. Permission and assistance to conduct the survey were secured from the local government unit referred to as barangay. Using maps provided by the local government unit, enumerators identified designated starting points and randomly selected an initial household in the vicinity of each point. If a household declined to participate, the next nearest household was approached. Thereafter, at least every twentieth household from the most recently interviewed household was approached for participation. Before the survey, enumerators explained its objectives and obtained informed consent from prospective respondents.

During the study's survey period, which largely coincides with the lockdown and mobility restrictions imposed by the government during the COVID-19 pandemic, cities in the NCR were subject to similar lockdown classifications as quarantine measures were implemented at the regional level. While minor differences in local enforcement and localized restrictions may have occurred, the overall policy environment remained largely uniform across cities, ensuring that pandemic-related conditions were comparable for all sampled households. The sampling design was intended to capture variation across key urban centers in the NCR. However, the sample is not population-weighted and does not adjust for differences in population size or density across cities. As such, the results should be interpreted as reflecting patterns within the sampled urban areas rather than being strictly representative of the entire NCR population and the Philippines.

#### 2.1.2 Climatological data

Daily mean temperature and relative humidity data were obtained from the Philippine Atmospheric, Geophysical and Astronomical Services Administration (PAGASA), which is the country's national weather agency. The Heat Index (HI) provides a measure of thermal stress and human discomfort during hot and humid conditions. It is computed using a nonlinear formula derived from regression analysis carried out by Rothfus (1990) that integrates air temperature and relative humidity. Higher HI values are associated with greater levels of discomfort and potential health risks particularly among vulnerable populations.

The monthly average HI was computed using daily data from January 2019 to April 2021, corresponding to the period of electricity consumption covered in the household survey. Data were sourced from three PAGASA weather stations located in the NCR: Port Area in Manila; Ninoy Aquino International Airport in Pasay City; and Science Garden in Quezon City. The 75th percentile daily HI values computed from 2001 to 2021 was used as the long-run or threshold HI, against which the actual daily HI values were compared. Cooling Degree Days (CDD), or the number of days when actual HI exceeds this threshold HI, was then used as another measure of climate variability.

#### 2.1.3 Empirical framework

The empirical analysis of the climate variability-electricity consumption link done in this study was based on the theoretical framework of Auffhammer & Mansur (2014) which posits that households maximize utility by

choosing the amount of electricity to consume and the number of appliances to purchase subject to their budget or income constraint. They note that climate variability affects energy consumption by changing energy consumption patterns in response to short run weather shocks and through long-run adaptation measures (such as the use of air-conditioners).

Monthly household electricity consumption data matched with corresponding climate variables was estimated using the model described in Eq. (1) below:

$$E_{ht} = \beta_0 + \beta_1 C_{ht} + \beta_2 Y_h + \beta_3 M_h + u_h + e_{ht} \quad (1)$$

where,  $E_{ht}$  is the electricity consumption of household  $h$  during period  $t$  (month-year),  $C_{ht}$  represents climate conditions which vary over time and across households, measured using data from the weather station nearest to the household's location.  $Y_h$  denotes household monthly income calculated as the median of the income bracket selected by the respondent in the survey.  $M_h$  is the vector of other household-level characteristics such as household size and ownership of electric appliances which together with income have been identified as key factors influencing residential electricity consumption (Brounen et al., 2012; Reiss & White, 2005). The unobserved time-invariant household-specific effect is denoted by  $u_h$ , while  $e_{ht}$  represents the idiosyncratic error term, capturing other factors not explained by the model.

We estimate both panel fixed effects (FE) and correlated random effects (Mundlak) specifications to account for the potential correlation between observed regressors and unobserved household characteristics. Panel analysis mitigates concerns related to non-random sampling for causal interpretation of within-sample relationships. These models rely on within-household variation over time and conditional comparisons across households, which do not require population weights for consistent estimation of regression coefficients under standard assumptions. The FE model controls for all time-invariant household characteristics by exploiting only within-household variation over time. In this specification, identification is based on how changes in climate conditions within a household are associated with changes in electricity consumption. As a result, time-invariant variables such as income and appliance ownership are not separately identified.

In order to retain these time-invariant variables while relaxing the strict assumptions of standard random effects, we also estimate a correlated random effects (Mundlak) model. The Mundlak approach is well established in the panel data literature as a flexible extension of standard random effects that relaxes the strict exogeneity assumption by allowing regressors to be correlated with unobserved heterogeneity through the inclusion of means of time-varying covariates (Mundlak, 1978; Schunck, 2013). Yang (2022) further shows that in one-way panel models, implementing the Mundlak device yields a correlated random effects estimator that is identical to the FE estimator. Following the implementation discussed in García Echeverri (2024), we augment Eq. (1) by including household-level averages of the time-varying climate variables, as shown in Eq. (2) below:

$$E_{ht} = \beta_0 + \beta_1 C_{ht} + \theta \bar{C}_h + \beta_2 Y_h + \beta_3 M_h + a_h + e_{ht} \quad (2)$$

where,  $\bar{C}_h$  denotes the household-specific average of the climate variables over time. Including these averages allows the unobserved household effect,  $a_h$ , to be correlated with the regressors, thereby relaxing the key assumption of standard random effects. In this framework, the coefficient,  $C_{ht}$ , capture the effect of short-term deviations in climate conditions, while  $\bar{C}_h$  reflect the average weather conditions associated with each household's location over time.

All models are estimated with standard errors clustered at the household level to account for serial correlation and heteroskedasticity. In addition, we include month dummy variables to control for recurring seasonal patterns in electricity consumption that may affect all households similarly. This helps ensure that the estimated effects of climate variables are not driven by systematic seasonal fluctuations.

Each household was assigned climate data from the PAGASA weather station located closest to its city of residence. Specifically, households in Quezon City and Caloocan (Northern District) were matched to the Science Garden station, while households in Manila City (Capital District) were assigned data from the Port Area station. For Makati City (Southern District), where households are located at approximately similar distances between two stations, climate variables were constructed as the simple average of observations from the Port Area and NAIA stations. This spatial matching approach helps ensure that household-level exposure more closely reflects localized weather conditions rather than relying on a single aggregate station for the entire region.

### 3. Results and Discussion

#### 3.1 Respondents' Profile

The summary profile of the sample of 500 respondents in the survey is presented in Table 1. The average household income is PhP 40,270 with a slightly higher standard deviation of PhP 40,712 indicating sufficiently

wide income variations in the sample. The monthly electricity consumption of the average household is 319 kWh. An average household has five members. As the survey was conducted during the COVID-19 pandemic when different forms of lockdowns and schemes such as work from home and online classes were being implemented to control the spread of the virus, an average of four out of five household members were at home even during the daytime. The average household has three electric fans and just one of each of the following electric appliances: air conditioner, refrigerator and freezer (combined), washing machine, computer, and television set.

**Table 1.** Profile of household respondents

| Variable                                                   | Income Group     |                 |                  |                   |
|------------------------------------------------------------|------------------|-----------------|------------------|-------------------|
|                                                            | All              | Low             | Middle           | High              |
| Number of observations                                     | 500              | 199             | 271              | 30                |
| Monthly household income (in Philippine Peso)              | 40270<br>(40712) | 16156<br>(8459) | 46089<br>(30021) | 147667<br>(57577) |
| Monthly electricity consumption (in Kilowatt-hour)         | 319<br>(317)     | 252<br>(276)    | 333<br>(274)     | 633<br>(600)      |
| Number of household members                                | 5.3<br>(2.5)     | 6.3<br>(2.8)    | 4.8<br>(2.1)     | 3.8<br>(1.8)      |
| Number of household members staying at home during daytime | 4.1<br>(2.4)     | 5.1<br>(2.7)    | 3.5<br>(1.9)     | 3.2<br>(1.7)      |
| Number of air conditioning units                           | 0.7<br>(1.2)     | 0.2<br>(0.5)    | 0.9<br>(1.2)     | 2.6<br>(1.9)      |
| Number of freezers and refrigerators                       | 0.9<br>(0.6)     | 0.7<br>(0.6)    | 1.0<br>(0.5)     | 1.4<br>(0.7)      |
| Number of electric fans                                    | 3.3<br>(1.7)     | 2.9<br>(1.4)    | 3.5<br>(1.8)     | 4.7<br>(2.2)      |
| Number of washing machines                                 | 0.9<br>(0.5)     | 0.8<br>(0.5)    | 0.9<br>(0.4)     | 1.1<br>(0.6)      |
| Number of computers                                        | 1.1<br>(1.5)     | 0.5<br>(0.8)    | 1.3<br>(1.4)     | 3.2<br>(2.3)      |
| Number of television sets                                  | 1.4<br>(0.9)     | 1.2<br>(0.7)    | 1.5<br>(0.9)     | 2.3<br>(1.3)      |

Notes: Authors' calculations. Standard deviations are indicated in parenthesis. Household income groups were identified using the monthly income ranges for a household of five computed by Albert et al. (2020) using 2018 prices. Adapting their classification, the corresponding monthly per capita income thresholds are as follows: low-income households earn less than PhP 4,383 per person per month; middle-income households earn between PhP 4,383 and PhP 26,297; and high-income households earn PhP 26,297 or more.

Source: Household survey.

Adapting the income group classification of Albert et al. (2020), the respondents were grouped into low-income (39.8% of the sample), middle-income (54.2%), and high-income (6.0%) groups (The distribution of the sample households by income class closely resembles the population distribution in the Philippines of 47.7% low-income, 50.2% middle income, and 2.1% high income. In the regions of Metro Manila, CALABARZON and Central Luzon, around three-fifths (60%) of the households belong to the middle-income group (Albert et al. (2020)). The average monthly household income among low-, middle-, and high-income households is PhP 16,156, PhP 46,089, and PhP 147,667, respectively.

### 3.2 Climate Variability

The Philippines has a tropical climate characterized by high temperatures, high humidity, and abundant rainfall. The average annual temperature in the country is 26.6 °C. With an average temperature of 25.5 °C, Jan is considered to be the coolest month. On the other hand, the warmest month is said to be May during which average temperature is 28.3 °C (PAGASA, n.d.). Temperature does not vary significantly with the latitude of any area in the Philippines. Only the altitude of the location results in some differences in temperature. Thus, there is essentially no difference in the mean temperature in different places in the country measured at or near sea level (PAGASA, n.d.). In NCR, the average elevation is 6 m, with minimum of -1 m and a maximum of 68 m (Manila Topographic Map, n.d.). Due to the high temperature and the numerous bodies of water in the Philippines, there is always high relative humidity, ranging from 71% in March to 85% in September. The combination of high temperature and high humidity results in warm temperature felt by the populace throughout the country, reaching uncomfortable levels during the months of March to May (PAGASA, n.d.).

Table 2 and Figure 2 show the monthly climatic variability in NCR during the survey period. The lowest average daily mean temperature of 26.96 °C during the survey period was recorded in Jan. The coldest months were from Dec to Feb. Temperature started to increase in Mar and peaked in May when average daily temperature reached 30.79 °C, before declining in the subsequent months. Relative humidity remained high throughout the year, ranging from 63 to 79%. Consequently, the HI was significantly higher than ambient temperature, ranging from



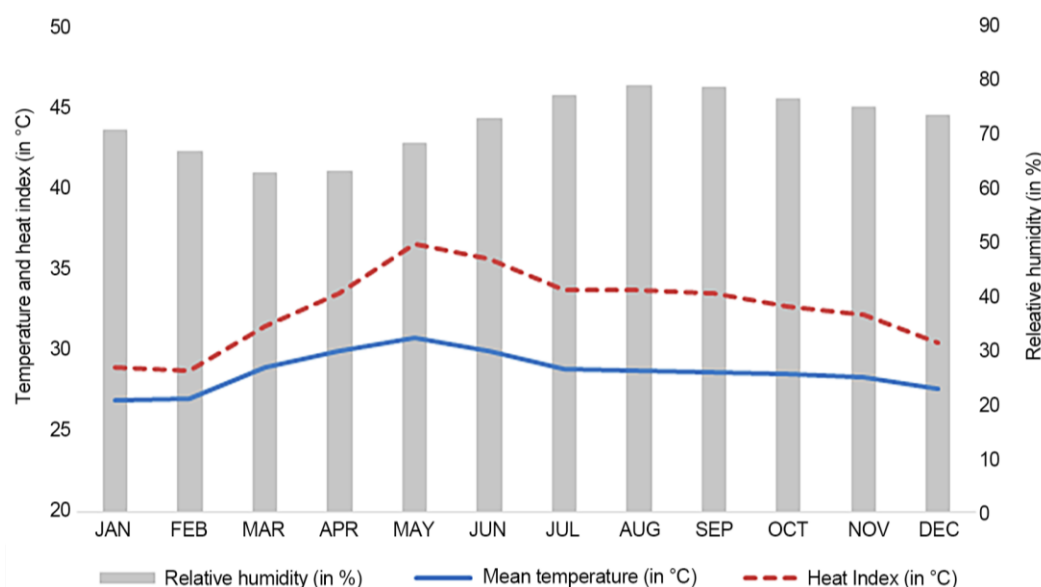
1.76 °C higher in Feb to 5.83 °C higher in May. Average daily HI was lowest in Feb at 28.81 °C and highest in May at 36.62 °C.

Using the 75th percentile of daily HI values from 2001 to 2021, the HI threshold was set at 34.82 °C, which falls within PAGASA’s “extreme caution” classification (PAGASA observes the following effect-based classification for the HI levels: less than 27 °C—not dangerous, 27–32 °C—caution, 33–41 °C—extreme caution, 42–51 °C—danger, and 52 °C and above—extreme danger (PAGASA, [n.d.](#))). Daily HI exceeded this threshold the most often during May.

**Table 2.** Monthly climate variability and electricity consumption

| Month | Average Daily Mean Temperature (°C) | Average Daily Relative Humidity (%) | Average Daily Heat Index (HI) (°C) | Average Number of Cooling Degree Days (CDD) | Average Electricity Consumption (kWh) |
|-------|-------------------------------------|-------------------------------------|------------------------------------|---------------------------------------------|---------------------------------------|
| Jan   | 26.96                               | 71.09                               | 28.99                              | 0                                           | 281                                   |
| Feb   | 27.05                               | 67.00                               | 28.81                              | 0                                           | 268                                   |
| Mar   | 28.93                               | 63.15                               | 31.55                              | 6                                           | 259                                   |
| Apr   | 30.01                               | 63.23                               | 33.61                              | 13                                          | 277                                   |
| May   | 30.79                               | 68.57                               | 36.62                              | 24                                          | 472                                   |
| Jun   | 29.98                               | 73.19                               | 35.66                              | 16                                          | 373                                   |
| Jul   | 28.91                               | 77.40                               | 33.74                              | 9                                           | 346                                   |
| Aug   | 28.74                               | 79.42                               | 33.81                              | 7                                           | 315                                   |
| Sep   | 28.70                               | 79.05                               | 33.61                              | 9                                           | 344                                   |
| Oct   | 28.52                               | 76.99                               | 32.78                              | 4                                           | 313                                   |
| Nov   | 28.37                               | 75.26                               | 32.19                              | 2                                           | 298                                   |
| Dec   | 27.64                               | 73.69                               | 30.47                              | 0                                           | 283                                   |

Note: Authors’ calculations. Climate data used were from three PAGASA weather stations located in the National Capital Region (NCR): Port Area in Manila; Ninoy Aquino International Airport in Pasay City; and Science Garden in Quezon City.  
Source: For climate variables, PAGASA ([n.d.](#)). Electricity consumption is from household survey.

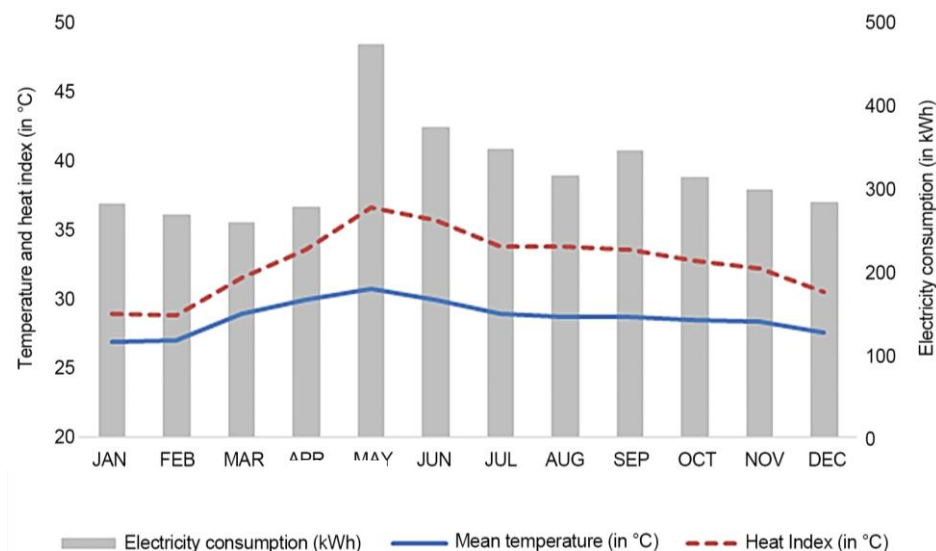


**Figure 2.** Average monthly climatic trends

Note: Authors’ calculations. Climate data used were from three PAGASA weather stations located in the National Capital Region (NCR): Port Area in Manila; Ninoy Aquino International Airport in Pasay City; and Science Garden in Quezon City.  
Source: PAGASA ([n.d.](#)).

### 3.3 Electricity Consumption and Climate Variability

Figure 3 plots monthly household electricity consumption with the climate variables, highlighting a positive correlation between electricity demand and seasonal variations in temperature and the HI. As temperature, HI, and CDD peak in May, so does household electricity use.

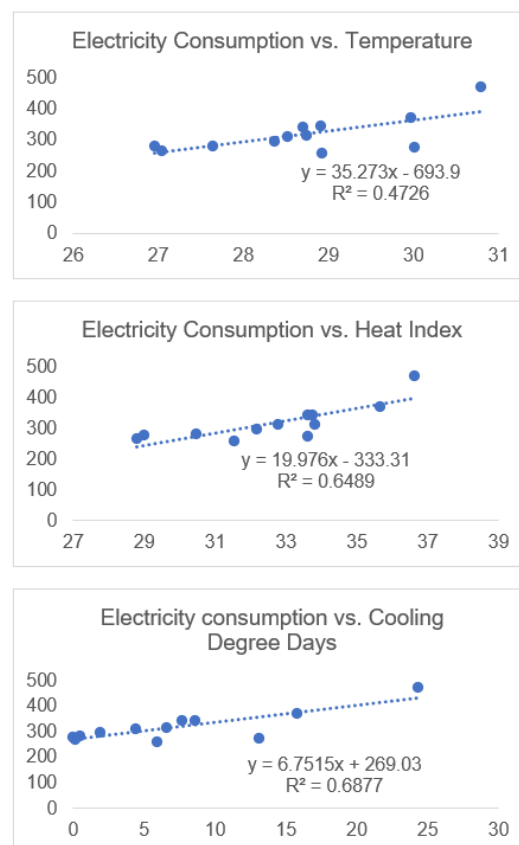


**Figure 3.** Average monthly electricity consumption and climatic trends

Note: Authors' calculations. Climate data used were from three PAGASA weather stations located in the National Capital Region (NCR): Port Area in Manila; Ninoy Aquino International Airport in Pasay City; and Science Garden in Quezon City.

Source: PAGASA (n.d.) and household survey.

Figure 4, which plots household monthly electricity consumption vis a vis the climate variables, reveals the positive linear correlation of electricity demand with climate. The  $R^2$  of the linear trends in Figure 4 indicate that variations in the HI and the CDD have stronger explanatory power for electricity use than temperature.



**Figure 4.** Linear trend of average monthly electricity consumption and climatic variability

Note: Authors' calculations. Climate data used were from three PAGASA weather stations located in the National Capital Region (NCR): Port Area in Manila; Ninoy Aquino International Airport in Pasay City; and Science Garden in Quezon City.

Source: PAGASA (n.d.) and household survey.

### 3.4 Determinants of Electricity Consumption

The results of the correlated random effects (Mundlak) regressions are summarized in Table 3. Models 1–3 pertain to the runs where the volume of monthly household electricity consumption in kWh is used as the dependent variable. The three models vary in terms of the explanatory climate variables used: Model 1 includes monthly average daily temperature and relative humidity as separate climate variables, Model 2 uses the monthly average daily HI, and Model 3 uses the number of CDD. The second set of regressions in Models 4–6 of Table 3 specifies monthly household electricity consumption and household income in logarithmic form. The regression runs yield robust results. The signs and magnitudes of the coefficients in the three alternative model specifications are essentially the same.

**Table 3.** Correlated random effects (Mundlak) regression results

| Explanatory Variables                | Dependent Variable: Average Monthly Electricity Consumption |                           |                       |                     |                     |                     |
|--------------------------------------|-------------------------------------------------------------|---------------------------|-----------------------|---------------------|---------------------|---------------------|
|                                      | Kilowatt-Hour                                               |                           |                       | Ln (Kilowatt-Hour)  |                     |                     |
|                                      | Model 1                                                     | Model 2                   | Model 3               | Model 4             | Model 5             | Model 6             |
| Income                               | 0.001*<br>(0.000)                                           | 0.001*<br>(0.000)         | 0.001*<br>(0.000)     | -<br>-              | -<br>-              | -<br>-              |
| LnIncome                             | -<br>-                                                      | -<br>-                    | -<br>-                | 0.194***<br>(0.037) | 0.184***<br>(0.037) | 0.186***<br>(0.037) |
| MembersHomeDay                       | 19.924***<br>(3.615)                                        | 19.944***<br>(3.606)      | 20.212***<br>(3.698)  | 0.062***<br>(0.010) | 0.062***<br>(0.010) | 0.062***<br>(0.010) |
| Aircon                               | 69.292***<br>(21.813)                                       | 69.917***<br>(21.839)     | 69.980***<br>(21.465) | 0.132***<br>(0.035) | 0.137***<br>(0.036) | 0.136***<br>(0.035) |
| Ref                                  | 48.094**<br>(20.009)                                        | 47.342**<br>(20.051)      | 47.018**<br>(20.150)  | 0.144***<br>(0.052) | 0.139***<br>(0.052) | 0.137***<br>(0.052) |
| Fan                                  | -16.566<br>(11.124)                                         | -16.286<br>(11.011)       | -15.762<br>(10.993)   | 0.007<br>(0.016)    | 0.010<br>(0.017)    | 0.012<br>(0.016)    |
| Wash                                 | 22.970<br>(20.512)                                          | 23.778<br>(20.361)        | 23.551<br>(20.353)    | 0.079<br>(0.053)    | 0.088*<br>(0.053)   | 0.086<br>(0.052)    |
| Computer                             | 45.355***<br>(15.726)                                       | 45.387***<br>(15.739)     | 46.208***<br>(15.847) | 0.051**<br>(0.022)  | 0.050**<br>(0.022)  | 0.050**<br>(0.022)  |
| TV                                   | 58.064***<br>(22.296)                                       | 59.058***<br>(22.098)     | 56.982***<br>(21.829) | 0.079**<br>(0.037)  | 0.089**<br>(0.037)  | 0.082**<br>(0.037)  |
| Temperature                          | 39.437***<br>(2.645)                                        | -<br>-                    | -<br>-                | 0.101***<br>(0.006) | -<br>-              | -<br>-              |
| Humidity                             | 3.281***<br>(0.321)                                         | -<br>-                    | -<br>-                | 0.011***<br>(0.001) | -<br>-              | -<br>-              |
| Temperature*Humidity                 | 0.880***<br>(0.267)                                         | -<br>-                    | -<br>-                | 0.002**<br>(0.001)  | -<br>-              | -<br>-              |
| Temperature_mean                     | 18.580<br>(46.966)                                          | -<br>-                    | -<br>-                | -0.007<br>(0.140)   | -<br>-              | -<br>-              |
| Humidity_mean                        | -3.015<br>(9.619)                                           | -<br>-                    | -<br>-                | -0.026<br>(0.029)   | -<br>-              | -<br>-              |
| Temperature_mean*Humidity_mean       | 5.679<br>(6.875)                                            | -<br>-                    | -<br>-                | 0.058**<br>(0.023)  | -<br>-              | -<br>-              |
| Heat Index (HI)                      | -<br>-                                                      | 19.017***<br>(1.280)      | -<br>-                | -<br>-              | 0.049***<br>(0.003) | -<br>-              |
| HeatIndex_mean                       | -<br>-                                                      | 16.985<br>(18.324)        | -<br>-                | -<br>-              | 0.056<br>(0.043)    | -<br>-              |
| CoolingDegreeDays                    | -<br>-                                                      | -<br>-                    | 6.379***<br>(0.418)   | -<br>-              | -<br>-              | 0.016***<br>(0.001) |
| CoolingDegreeDays_mean               | -<br>-                                                      | -<br>-                    | 4.635<br>(7.183)      | -<br>-              | -<br>-              | 0.043*<br>(0.025)   |
| Constant                             | -39.521<br>(34.995)                                         | -1,198.718**<br>(595.065) | -116.987*<br>(61.339) | 2.659***<br>(0.351) | -0.656<br>(1.466)   | 2.275***<br>(0.404) |
| Observations                         | 5,679                                                       | 5,679                     | 5,679                 | 5,679               | 5,679               | 5,679               |
| R <sup>2</sup> (overall)             | 0.4127                                                      | 0.4132                    | 0.4174                | 0.4303              | 0.4257              | 0.4320              |
| chi2                                 | 466.18                                                      | 438.25                    | 430.89                | 1007.41             | 957.67              | 1020.90             |
| Prob > chi2                          | 0.000                                                       | 0.000                     | 0.000                 | 0.000               | 0.000               | 0.000               |
| Number of households                 | 496                                                         | 496                       | 496                   | 496                 | 496                 | 496                 |
| Month fixed effects (FE)             | Yes                                                         | Yes                       | Yes                   | Yes                 | Yes                 | Yes                 |
| Robust standard errors               | Yes                                                         | Yes                       | Yes                   | Yes                 | Yes                 | Yes                 |
| Mean variance inflation factor (VIF) | 2.23                                                        | 1.51                      | 1.51                  | 2.20                | 1.48                | 1.48                |
| Maximum VIF                          | 9.50                                                        | 2.23                      | 2.22                  | 9.46                | 1.95                | 1.95                |

Note: Robust standard errors are in parentheses; \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.1$ ; “-” = Variable excluded from the model.



In accordance with economic theory, income has a significant positive impact on electricity consumption. All other factors held constant, a PhP 1,000 increase in household income is associated with a 1 kWh increase in monthly electricity consumption. The coefficient of income in the double-log form, 0.18 to 0.19, is less than one indicating income-inelastic electricity demand, as found in past studies (Danao, 2001; Danao & Ducanes, 2016). Another household characteristic that turns out to be significant is the number of household members that stay at home during the daytime. In the log specification, each additional household member staying at home during the daytime increases electricity consumption by approximately 6.2%, reflecting higher appliance usage when more individuals are present at home during the day. Noting that the household survey was conducted during the COVID-19 pandemic when lockdowns and work from home and online classes were done to control the spread of the virus, household size and the number of household members staying at home during the daytime were closely correlated. The latter, which yields better overall regression results, was used. As expected, an additional member present during the day raises monthly electricity consumption by about 20 kWh, holding other factors constant.

Of the electrical appliances included in the regressions, air conditioners, refrigerators/freezers, computers and television sets have significant positive coefficients. Air-conditioning units, the most energy-intensive appliance, has the highest coefficient. On average, one additional air-conditioning unit increases monthly electricity consumption by about 69 to 70 kWh, or roughly 13–14% in the log specification. Another energy-intensive appliance, refrigerator/freezer, has a lower but nevertheless high coefficient. An additional refrigerator/freezer unit increases monthly electricity consumption by about 47 to 48 kWh. Television and computer also have substantial impact on electricity consumption, contributing approximately 57–59 kWh (8–9%) and 45–46 kWh (5%), respectively. This can be attributed to the extended time usage on a daily basis of these appliances in Philippine households. For instance, watching television is a favorite and affordable pastime among households.

Most household-average climate variables are statistically insignificant. This suggests that differences across households in their typical climate conditions do not play a major role in explaining variation in electricity consumption once household characteristics and monthly weather fluctuations are controlled for. Instead, electricity consumption is primarily driven by within-household responses to changes in weather over time rather than differences in average climate exposure across locations.

The FE regression results are reported in Table 4. By construction, the specification controls for all time-invariant household characteristics and as a result, the model estimates only the coefficients of time-varying climate variables. Importantly, the FE estimates are largely consistent in magnitude and sign with those obtained from the Mundlak specification, suggesting that the main results are robust to alternative modeling approaches and are not driven by unobserved time-invariant household heterogeneity.

**Table 4.** Panel fixed effects (FE) regression results

| Explanatory Variables  | Dependent Variable: Average Monthly Electricity Consumption |                         |                       |                     |                     |                     |
|------------------------|-------------------------------------------------------------|-------------------------|-----------------------|---------------------|---------------------|---------------------|
|                        | Kilowatt-Hour                                               |                         |                       | Ln (Kilowatt-Hour)  |                     |                     |
|                        | Model 1                                                     | Model 2                 | Model 3               | Model 4             | Model 5             | Model 6             |
| Temperature            | 39.437***<br>(2.645)                                        | -                       | -                     | 0.101***<br>(0.006) | -                   | -                   |
| Humidity               | 3.281***<br>(0.321)                                         | -                       | -                     | 0.011***<br>(0.001) | -                   | -                   |
| Temperature*Humidity   | 0.880***<br>(0.266)                                         | -                       | -                     | 0.002**<br>(0.001)  | -                   | -                   |
| Heat Index (HI)        | -                                                           | 19.017***<br>(1.278)    | -                     | -                   | 0.049***<br>(0.003) | -                   |
| CoolingDegreeDays      | -                                                           | -                       | 6.379***<br>(0.414)   | -                   | -                   | 0.016***<br>(0.001) |
| Constant               | 310.951***<br>(0.999)                                       | -295.654***<br>(41.300) | 270.933***<br>(3.139) | 5.433***<br>(0.002) | 3.885***<br>(0.086) | 5.335***<br>(0.006) |
| Observations           | 5,679                                                       | 5,679                   | 5,679                 | 5,679               | 5,679               | 5,679               |
| R-squared (overall)    | 0.0202                                                      | 0.0211                  | 0.0274                | 0.0236              | 0.0240              | 0.0293              |
| chi2                   | 74.94                                                       | 221.43                  | 233.64                | 115.22              | 336.94              | 442.86              |
| Prob > chi2            | 0.000                                                       | 0.000                   | 0.000                 | 0.000               | 0.000               | 0.000               |
| Number of households   | 496                                                         | 496                     | 496                   | 496                 | 496                 | 496                 |
| Robust standard errors | Yes                                                         | Yes                     | Yes                   | Yes                 | Yes                 | Yes                 |

Note: Robust standard errors are in parentheses; \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.1$ ; “-” = Variable excluded from the model.

Temperature and humidity both exert positive and statistically significant effects on electricity consumption. Moreover, the interaction term further indicates that their effects are interdependent. Higher humidity amplifies the effect of temperature on electricity consumption, and vice versa. Specifically, in Model 1, the marginal effect of temperature is given by  $39.437 + 0.880 \times \text{Humidity}$ , implying that the impact of temperature on electricity consumption increases as humidity rises. Likewise, the marginal effect of humidity is  $3.281 + 0.880 \times \text{Temperature}$ ,

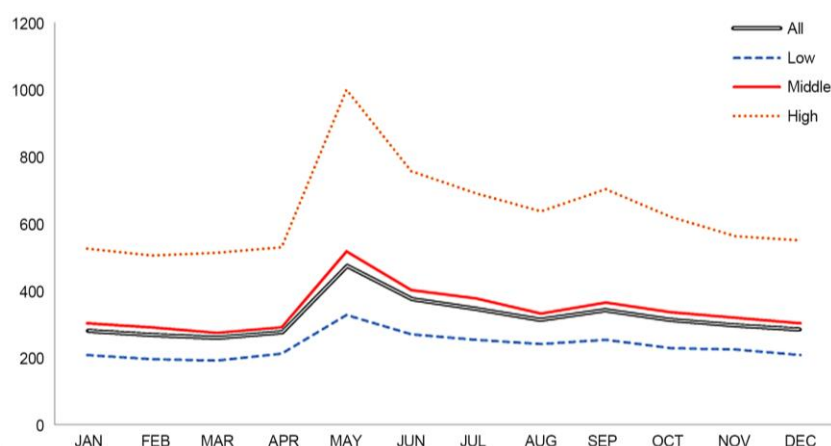
suggesting that the effect of humidity intensifies at higher temperatures. Overall, these results point to a mutually reinforcing relationship between temperature and humidity in shaping electricity demand. Electricity consumption increases disproportionately when high temperature coincides with high humidity, consistent with higher perceived heat stress and increased reliance on cooling appliances.

HI, a composite climate index of temperature and relative humidity, has a strong positive effect on electricity consumption by approximately 19 kWh (4.9% in logs), confirming that perceived heat conditions are strongly associated with increased energy demand. Similarly, CDD are positively and significantly associated with electricity consumption, with coefficients of 6.4 kWh (1.6% in logs), indicating higher electricity use during periods of greater cooling demand.

The results are consistent with a growing body of international evidence documenting a positive relationship between warmer climate and electricity consumption. Ang et al. (2017), studying Singapore and Hong Kong, find that a 1 °C increase in temperature would raise total annual electricity consumption by approximately 3%–4% in Singapore and 4%–5% in Hong Kong, with the residential sector exhibiting the strongest response among end-users. Similarly, Avordeh et al. (2021), examining the Greater Accra Region in Ghana, estimate a log-linear electricity demand model and find that residential electricity consumption increases by up to 3.1% under higher temperature conditions, highlighting the sensitivity of demand to climate variability in developing country contexts. In the Philippine setting, Dacuycuy (2019) uses household-level data from the 2011 Household Energy Consumption Survey and finds that electricity consumption increases substantially with higher heat exposure. Specifically, household electricity use rises by approximately 33 to 43 kWh when the HI increases by 1 °C, and by 68 to 88 kWh when it increases by 2 °C above normal levels.

### 3.5 Climate Sensitivity of Electricity Consumption, by Household Income Groups

Past studies suggest that the sensitivity of electricity consumption to climate varies across income groups. Figure 5 reveals that not only is the electricity use of the high-income group much higher than the two lower income groups, high-income households' electricity consumption, on average, peaks much more steeply than the low- and middle-income groups during the extreme climate month of May.



**Figure 5.** Average monthly electricity consumption by income groups

Note: Authors' calculations.

Source: Household survey.

Table 5 presents the coefficients of the climate variables when correlated random effects (Mundlak) regressions (with electricity consumption volume in kWh as dependent variable) are estimated separately for the three income groups. The results confirm the observation in Figure 5. The magnitudes of the coefficients of the climate variables increase across the income groups. Nevertheless, future research using more balanced and representative income-group samples would be useful to validate and extend these findings.

Similar to the results for all households, both temperature and humidity exert a positive and statistically significant effect on electricity consumption across income groups. Among low-income households, the interaction coefficient of 0.984 implies that each additional unit of humidity increases the marginal effect of temperature on electricity consumption by 0.984 kWh. For middle-income households, the corresponding interaction effect is weaker at 0.535. Generally, the coefficient magnitudes increase with income. The coefficients for the middle-income group are less than twice those of the low-income group, whereas those for the high-income group exceed twice those of the middle-income group. This suggests that electricity demand of higher income households is more sensitive to climate fluctuations. Conversely, the weaker responsiveness of electricity demand among lower-

income households may reflect constraints in adapting to climate variability, suggesting equity concerns associated with the non-market costs of climate change consistent with the findings of Li et al. (2019).

**Table 5.** Correlated random effects (Mundlak) regression results by income groups

| Model                                                                       | Climate Variable       | All Households       | Low-Income           | Middle-Income        | High-Income           |
|-----------------------------------------------------------------------------|------------------------|----------------------|----------------------|----------------------|-----------------------|
| Dependent variable: Average monthly electricity consumption (Kilowatt-hour) |                        |                      |                      |                      |                       |
| Model 1                                                                     | Temperature            | 39.437***<br>(2.645) | 24.968***<br>(3.012) | 43.561***<br>(3.570) | 97.705***<br>(19.287) |
|                                                                             | Humidity               | 3.281***<br>(0.321)  | 2.288***<br>(0.448)  | 3.730***<br>(0.430)  | 7.321***<br>(2.319)   |
|                                                                             | Temperature*Humidity   | 0.880***<br>(0.267)  | 0.984**<br>(0.463)   | 0.535*<br>(0.300)    | 0.982<br>(1.446)      |
|                                                                             | HeatIndex              | 19.017***<br>(1.280) | 11.915***<br>(1.473) | 21.131***<br>(1.738) | 47.770***<br>(9.345)  |
| Model 2                                                                     |                        |                      |                      |                      |                       |
| Model 3                                                                     | CoolingDegreeDays      | 6.379***<br>(0.418)  | 4.016***<br>(0.432)  | 7.099***<br>(0.584)  | 15.291***<br>(2.976)  |
|                                                                             | Number of observations | 5,679                | 2,235                | 3,095                | 349                   |
|                                                                             | Number of households   | 496                  | 196                  | 270                  | 30                    |

Note: Robust standard errors are in parentheses. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.1$ . Household income groups were identified using the monthly income ranges for a household of five computed by Albert et al. (2020) using 2018 prices. Adapting their classification, the corresponding monthly per capita income thresholds are as follows: low-income households earn less than PhP 4,383 per person per month; middle-income households earn between PhP 4,383 and PhP 26,297; and high-income households earn PhP 26,297 or more. Results for the high-income group are exploratory due to the relatively smaller sample size.

The results highlight important dimensions of energy inequality, as households with higher incomes and greater ownership of energy-intensive appliances (such as air conditioners and refrigerators) exhibit substantially higher electricity consumption and are better able to respond to heat stress. This indicates uneven access to cooling services, where lower-income households may face constraints in maintaining thermal comfort during hot and humid periods. As a result, the ability to adapt to climate variability is not evenly distributed across households. From a policy perspective, these findings support targeted interventions such as minimum energy efficiency standards for cooling appliances, appliance upgrading or subsidy programs for low-income households, and demand-side management measures during peak demand periods. These interventions can help reduce both energy burden and exposure to extreme heat. More broadly, the strong sensitivity of electricity demand to temperature and humidity underscores the climate vulnerability of poor, urban households in tropical settings, particularly those with limited adaptive capacity.

#### 4. Conclusions

This study examined the effect of climate variability on monthly residential electricity consumption using novel household-level survey data. By combining detailed micro-level information on household characteristics and appliance ownership with monthly electricity usage, the study provides a more granular assessment of residential energy demand than is typically possible with aggregate or utility-level data. The study is also among the limited empirical analyses of residential electricity demand in the country, particularly using household-level panel data and climate variation. In addition, we improve the measurement of climate exposure by matching households to the nearest PAGASA weather stations based on location, ensuring more precise spatial alignment of weather conditions. This reduces measurement error in climate variables and strengthens the credibility of the estimated effects.

As estimates of the impact of climate change on household energy demand are typically used to predict the cost of climate change adaptation, they provide essential inputs in the formulation and implementation of climate mitigation and adaptation policies. The study found that among households in selected cities during the COVID-19 pandemic, residential electricity demand is significantly responsive to seasonal or monthly fluctuations in temperature and the HI. Electricity consumption increases dramatically and reaches peak levels during the extremely hot days of the month of May. The correlated-random effects (Mundlak) regression results reveal statistically significant and substantial sensitivity of household electricity demand to the rise in temperature and HI. A 1 °C increase in average daily temperature increases monthly electricity consumption by about 39 kWh; while a 1 °C increase in the HI, a composite climate index of temperature and relative humidity, is associated with about 19 kWh increase in monthly electricity consumption. In terms of CDD, one more day of the HI surpassing the long-run threshold of 34.82 °C increases monthly electricity consumption by about 6 kWh. These findings imply that climate fluctuations beyond the normal temperature and HI induce changes in NCR households' behavior and routines in the use of cooling appliances. The estimated positive relationship between temperature, humidity, and electricity consumption is consistent with increased demand for cooling services during periods of higher thermal stress. Overall, the results suggest that households adjust their electricity consumption in response

to perceived heat conditions through a range of potential cooling-related behaviors.

The study also found that electricity consumption of lower income households is less sensitive to climate fluctuations. Lower income households are not able to fully adjust their electricity consumption in accordance with extreme temperature and HI even when they are unbearable due to insufficient budget to accommodate a higher electricity bill and to purchase better cooling appliances. This indicates non-market costs of climate change that inequitably burden lower income households.

On the other hand, the higher climate sensitivity of electricity demand of high-income households indicates the viability of promoting adoption of energy-saving cooling technologies and roof-top solar power by high-income households who can afford the new and cleaner energy technologies. These findings provide important inputs in energy sector planning and climate change mitigation and adaptation program development.

### Author Contributions

Conceptualization, R.P.-T.; methodology, R.P.-T.; software, R.P.-T. and G.P.; validation, R.P.-T.; formal analysis, R.P.-T. and G.P.; investigation, R.P.-T. and G.P.; resources, R.P.-T.; data curation, R.P.-T. and G.P.; writing—original draft preparation, R.P.-T. and G.P.; writing—review and editing, R.P.-T. and G.P.; visualization, R.P.-T. and G.P.; supervision, R.P.-T.; project administration, R.P.-T.; funding acquisition, R.P.-T. All authors have read and agreed to the published version of the manuscript.

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### Informed Consent Statement

Informed consent was obtained from all subjects involved in the study.

### Data Availability

The data used to support the research findings are available from the corresponding author upon request.

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### Conflicts of Interest

The authors declare no conflicts of interest.

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## Nomenclature

|     |                 |
|-----|-----------------|
| °C  | degree Celsius  |
| kWh | Kilowatt-hour   |
| PhP | Philippine Peso |